DOMESTIC AIRLINES — TRAINING

Inoperative Ground Equipment

Course handbook

How failed or downgraded ground equipment — ILS components, navaids, RVR systems and lighting — affects approach minima, and how to apply the OM A table.

3 lessons · 1 modules · about 0.5 hours

Training material only. It does not replace the Aircraft Flight Manual, FCOM, QRH, MEL or the Domestic Airlines Operations Manual, which always take precedence.

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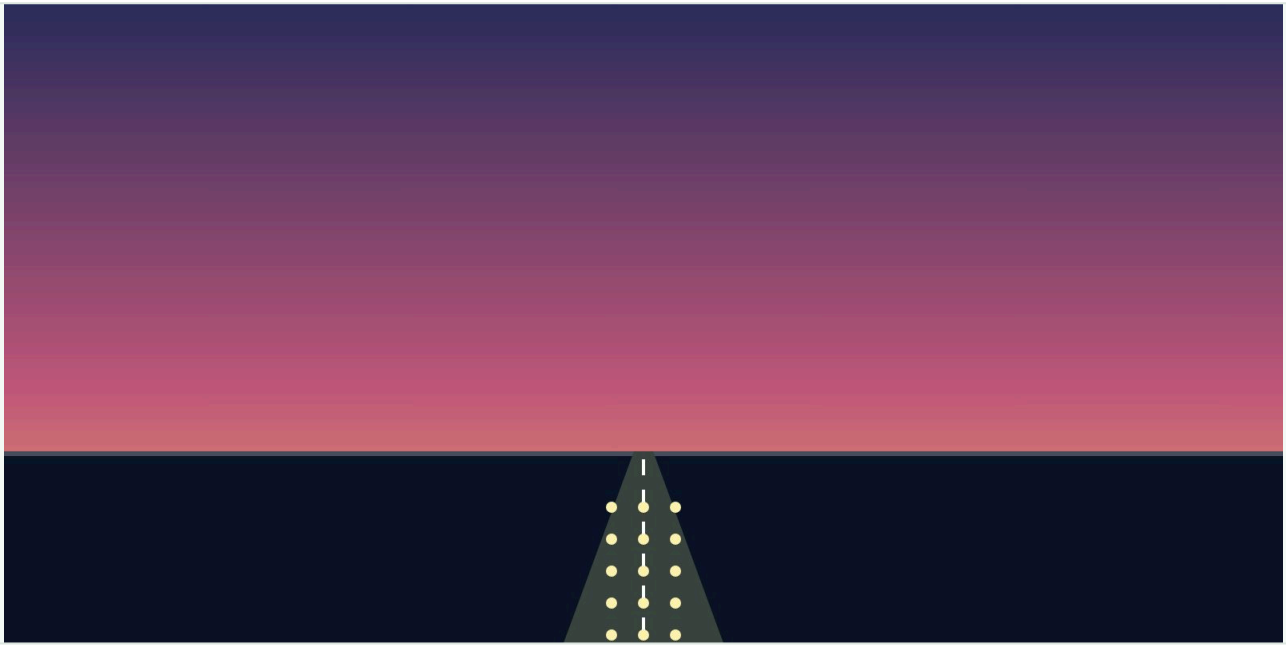
MODULE 1 · INOPERATIVE GROUND EQUIPMENT

Lesson 01: Why Ground Equipment Matters

Lesson 1 of 3 · about 5 min

LEARNING OBJECTIVES

- Explain why published minima assume serviceable ground equipment.
- Identify how failures are notified.
- Describe the possible effects on an approach.



Minima assume serviceable aids and lighting.

Minima assume serviceable equipment

Aerodrome operating minima are established for a runway with its navigation aids, lighting and RVR equipment serviceable (EASA CAT.OP.MPA.110; ICAO Doc 9365). When an item fails or is downgraded, the minima may have to be **increased**, or the approach may **not be available**.

How you find out

- **NOTAM** — planned or known outages (e.g. ILS glide path unserviceable, approach lights out of service).
- **ATIS and ATC** — recent failures.
- Flight deck indications — loss of an ILS signal, flags, or visible lighting failures.

Possible effects

- **No effect** — for example a failure that has an equivalent replacement (a marker replaced by a DME distance or another fix).
- **Higher minima** — for example an increased RVR when approach lights are out.

- **Downgrade** — for example from CAT II to CAT I, or from a 3D to a 2D approach (ILS to LOC only).
- **Approach not available**.

COMPANY TABLE

The effect of each failed or downgraded item on the minima is given in the **"failed or downgraded equipment" table of the Domestic Airlines OM A**. Use that table — this course explains how it works, not its values.

KEY TAKEAWAYS

- Minima assume serviceable ground aids, lighting and RVR equipment.
- Failures come from NOTAM, ATIS / ATC or flight deck indications.
- Effects: none, higher minima, downgrade, or no approach.

KNOWLEDGE CHECK

The ILS glide path is out of service. What is the usual effect?

A. No effect

B. The approach becomes a localiser-only (2D) approach with higher minima

C. CAT III becomes available

D. The runway closes

Answer: B. Without the glide path the ILS becomes a LOC-only 2D approach.

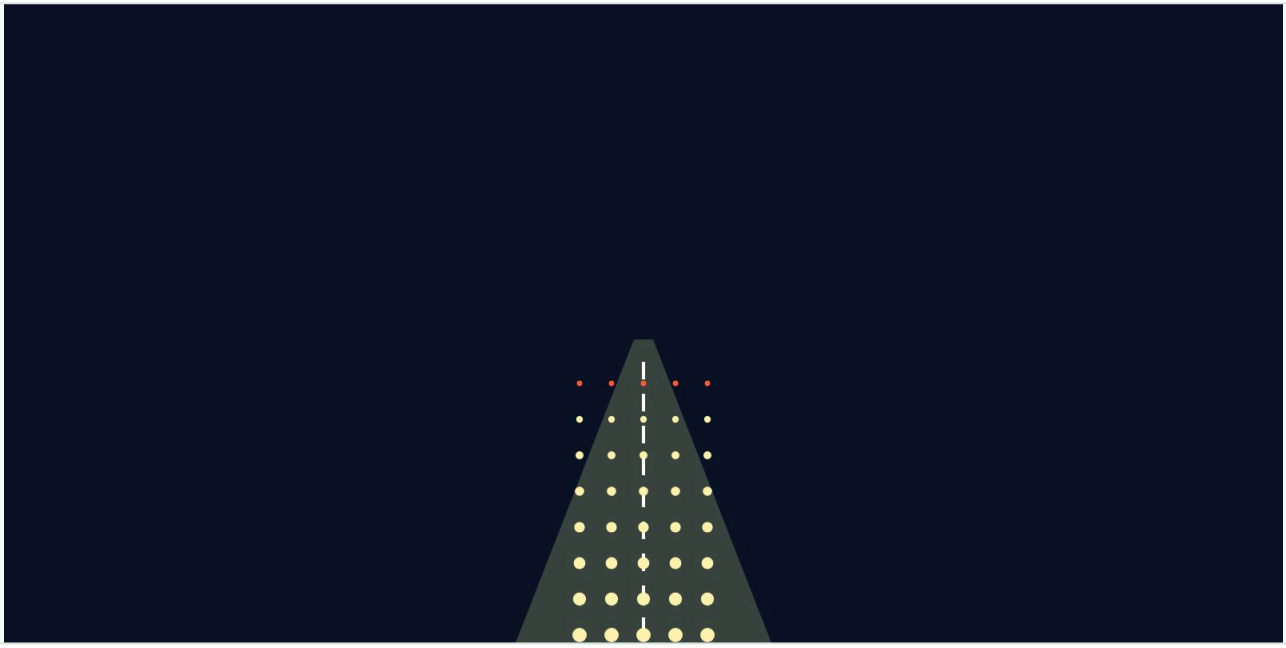
MODULE 1 · INOPERATIVE GROUND EQUIPMENT

Lesson 02: The Equipment and the Table

Lesson 2 of 3 · about 7 min

LEARNING OBJECTIVES

- Identify the ground equipment covered by the table.
- Understand approach lighting classes.
- Read the failed equipment table correctly.



The lighting class sets the RVR minimum.

Equipment covered

- **ILS / GLS components** — localiser, glide path, standby transmitters, outer and middle markers or the DME replacing them.
- **Navigation aids** for non-precision approaches — VOR, NDB, DME.
- **RVR assessment systems** — touchdown zone, mid-point and stop-end sensors.
- **Lighting** — approach lights, runway edge, threshold and end lights, runway centreline lights, touchdown zone lights, and standby power for lighting.

Approach lighting classes (EASA)

Class	Length of the approach lighting
FALS — full approach lighting	720 m or more
IALS — intermediate	420 m to 719 m
BALS — basic	210 m to 419 m
NALS — no approach lighting	less than 210 m, or none

The minimum RVR for an approach depends on the lighting class. When part of the approach lighting fails, the class can change — and the RVR minimum increases.

Reading the table

1. Find the failed or downgraded item.
2. Find the column for your type of approach (e.g. CAT II/III, CAT I, 2D).
3. Read the effect: no effect, a modified condition, an increased RVR, or “not allowed”.
4. Apply only **one entry per failed item**; if several items have failed, apply the most restrictive result, and check the notes in the table.

COMPANY TABLE

The effect of each failed or downgraded item on the minima is given in the **“failed or downgraded equipment” table of the Domestic Airlines OM A**. Use that table — this course explains how it works, not its values.

KEY TAKEAWAYS

- Covered: ILS components, navaids, RVR systems, lighting.
- Lighting classes FALS / IALS / BALS / NALS set the RVR minimum.
- Read the item, the approach type column and the effect; apply the most restrictive result for multiple failures.

KNOWLEDGE CHECK

Approach lights 450 m long belong to which class?

A. FALS

B. IALS

C. BALS

D. NALS

Answer: B. IALS: 420 m to 719 m.

MODULE 1 · INOPERATIVE GROUND EQUIPMENT

Lesson 03: Applying the Allowances in Flight

Lesson 3 of 3 · about 5 min

LEARNING OBJECTIVES

- Apply the table during planning and in flight.
- Brief the revised minima.
- Handle a failure during the approach.



Recalculate and brief the minima with the latest ATIS.

Planning

- Check the NOTAMs for the destination and alternates; apply the table to calculate the revised minima.
- Check that the weather allows the revised minima, with the planning minima for alternates.

In flight

- Get the ATIS, confirm the status of aids and lighting, recalculate the minima and **brief** them.
- If a failure is reported **during the approach**, assess the effect: on a precision approach, a failure of the guidance usually requires a go-around unless the required visual reference is already established, as your procedure states.
- Do not improvise minima — if in doubt, use the more restrictive value or go around.

Reporting

Report any observed failure of aids or lighting to ATC; it may not have been detected on the ground.

COMPANY TABLE

The effect of each failed or downgraded item on the minima is given in the **"failed or downgraded equipment" table of the Domestic Airlines OM A**. Use that table — this course explains how it works, not its values.

KEY TAKEAWAYS

- Apply the table at planning and again in flight with the latest ATIS.
- Brief the revised minima.
- If in doubt, use the more restrictive value or go around; report failures to ATC.

KNOWLEDGE CHECK

During an ILS approach, ATC reports the glide path unserviceable above the minimum. What do you do?

A. Continue to the DA

B. Assess and normally go around or change to the LOC approach and minima as the procedure states

C. Descend faster

D. Ignore it

Answer: B. A failure of guidance on the approach usually requires a go-around or a change to the appropriate approach and minima.

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